

# RTL Design Sherpa

**RTL AMBA CXS Module Reference — Rev 0.90**

**October 2026**

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# 1 CXS (Credited eXtensible Stream) Modules

**Location:** `rtl/amba/cxs/` **Specification:** Arm *AMBA CXS Protocol Specification*, ARM IHI 0079, Issues A, B, C and D **Status:** New (2026-10-15); GATE/FUNC/FULL green

## 1.1 Overview

CXS is Arm's point-to-point packet stream protocol for wide links. Its main use is between an on-chip interconnect and a PCIe controller carrying CCIX or CXL. Data moves in one direction, one *flit* (one cycle of `CXSDATA`) at a time. A flit can hold several packets, and a packet can span several flits.

CXS differs from AXI-Stream in one fundamental way: **there is no READY.** The receiver grants credits on `CXSCRDGNT`, one per cycle. The transmitter spends one credit per flit, and with link control it can hand a credit back on `CXSCRDRTN`. Every credit the receiver grants is a promise of buffer space for one flit, so the receiver is never back-pressured on the wire.

This directory holds one family covering **every published issue**. Each issue is a strict superset of the one before it. An issue is therefore a set of legal property values, not a different protocol, and the `CXS_ISSUE` parameter selects which set is checked at elaboration.

| Issue | Date | What it adds |
|---|---|---|
| A | Mar 2018 | First release: `CXSDATAFLITWIDTH` 256/512/1024, `CXSMAXPKTPERFLIT` 2..4, up to 15 credits, `CXSCHECKTYPE`, `CXSLINKCONTROL` |
| B | Dec 2020 | Multiple protocol streams: `CXSLAST` (`CXS_LAST`), `CXSPRCLTYPE` (`CXS_PROTOCOL_TYPE`), and their check signals |
| C | Dec 2023 | One packet per flit (`CXSMAXPKTPERFLIT` = 1, no `CXSCNTL`, any width 8..2048), `CXS_MAX_CREDIT` up to 63, clock and reset |

| Issue | Date | What it adds |
|---|---|---|
| | | defined |
| D | Sep 2025 | 2048-bit flits with several packets, `CXS_START_ALIGNMENT` = 4, `CXSSRCID`/`CXSTGTID` (0..8 bits), `CXSUSER` (0..128 bits), CXS-Lite named |

## 1.2 Module Categories

| Module | Role | Page |
|---|---|---|
| `cxs_master` | Transmitter: FUB valid/ready flits in, credited CXS link out | cxs_master.md |
| `cxs_slave` | Receiver: credited CXS link in, FUB valid/ready flits out; the receive FIFO is the credit pool | cxs_slave.md |
| `cxs_master_cg` | `cxs_master` behind `amba_clock_gate_ctrl`; gates only in link STOP | cxs_master_cg.md |
| `cxs_slave_cg` | `cxs_slave` behind `amba_clock_gate_ctrl`; a rising `CXSACTIVEREQ` wakes it | cxs_slave_cg.md |
| `cxs_pkg` (`rtl/amba/includes/`) | Widths, `CXSCNTL` field offsets, per-issue legality (`config_error`) | this page |
| `cxs_cntl_track` | Packet-open tracking and `CXSCNTL` format checks, shared by both endpoints | this page |
| `cxs_odd_byte_parity` | `Odd_Byte_Parity` generate/check for one signal of any width | this page |

| Module | Role | Page |
|---|---|---|
| cxs_stream_key | The tracked part of a flit's stream identity: {PRCLTYPE, SRCID, TGTID} low bits | this page |
| cxs_link_snoop | Passive model of a link from its wires alone: credits in flight, link state, credit/link/per-stream framing checks (feeds cxs_intf_observer) | this page |
| cxs_master_monlite / cxs_slave_monlite | The endpoints with cxs_monitor_lite tapping their FUB handshake and status; packets tagged PROTOCOL_CXS | cxs_monitor_lite.md |
| axis_to_cxs / cxs_to_axis | AXI-Stream beats to CXS flits and back (one packet per TLAST run, TSTRB carried in CXSUSER). They live in projects/components/utility-ip/converters/rtl/ | Converters MAS 3.17 (17_axis_to_cxs.md) |

The monitor wrappers have no _cg variants (clock gating is an FPGA-only concern here). Both endpoints export credit_starved for the monitor: on the transmitter, a flit is waiting in RUN with no credit; on the receiver, no credit can be granted because the FIFO holds every slot.

## 1.3 Properties and Parameters

Every endpoint takes the same property set. Parameter names follow the spec property they implement.

| Spec property | Parameter | Values | Issue |
|---|---|---|---|
| (issue) | CXS_ISSUE | 1..4 = A..D (cxs_pkg::CXS_ISSUE_A..D) | - |

| Spec property | Parameter | Values | Issue |
|---|---|---|---|
| CXSDATAFLITWIDTH | CXS_DATA_WIDTH | 256/512/1024; 2048 with several packets (D); 8..2048 by 8 with one packet (C) | A |
| CXSMAXPKTPERFLIT | CXS_MAX_PKT_PER_FLIT | 2..4; 1 from C | A |
| CXS_START_ALIGNMENT | CXS_START_ALIGN | 16; 4 from D | D |
| CXS_LAST | CXS_HAS_LAST | 0/1 | B |
| CXS_PROTOCOL_TYPE | CXS_HAS_PRCLTYPE | 0/1 | B |
| CXS_SRCID_WIDTH / CXS_TGTID_WIDTH | same names | 0..8 | D |
| CXS_USER_WIDTH | same name | 0..128 | D |
| CXS_MAX_CREDIT | same name | 1..15; 1..63 from C | A |
| CXSLINKCONTROL | CXS_LINK_CTRL | 1 = Explicit_Credit_Return, 0 = None | A |
| CXSCHECKTYPE | CXS_CHECK | 1 = Odd_Byte_Parity, 0 = None | A |
| (beyond spec) | CXS_CREDIT_BEYOND_SPEC | 1 = CXS_MAX_CREDIT may exceed the cap (rule 12 skipped) | - |
| (stream tracking) | CXS_STREAM_PRCL_BITS / CXS_STREAM_SRCID_BITS / CXS_STREAM_TGTID_BITS | low bits of each field that tell interleaved streams apart; 0/0/0 = one stream | B |

CXSCONTINUOUSDATA and CXSERRORFULLPKT constrain what the *source of packets* does, so they are not parameters of these endpoints. Both endpoints carry flits unchanged. The transmitter never deactivates the link while a packet is open, which CXSCONTINUOUSDATA requires.

### 1.3.1 Configuration checks

`cxs_pkg::config_error()` returns 0 for a legal property set, or the number of the first rule broken. Each endpoint calls it from an elaboration-time `param_check` and stops with `$fatal` naming the rule. The framework's `CXSConfig.config_error()` implements the same rules in the same order, so a rule number means the same thing on both sides.

| Rule | Requirement |
|---|---|
| 1 | Issue is A..D |
| 2 | `CXSMAXPKTPERFLIT` is 1..4 |
| 3 | One packet per flit needs Issue C or later |
| 4 | With one packet per flit, the width is 8..2048 and a multiple of 8 |
| 5 | With one packet per flit, `CXS_LAST` and `CXS_PROTOCOL_TYPE` are false |
| 6 | With several packets per flit, the width is 256/512/1024, plus 2048 from Issue D |
| 7 | A 256-bit flit carries at most 2 packets, unless start alignment is 4 (Issue D) |
| 8 | Start alignment is 4 or 16 |
| 9 | Start alignment 4 needs Issue D |
| 10 | `CXSLAST` / `CXSPRCLTYPE` need Issue B or later |
| 11 | `CXS_MAX_CREDIT` is at least 1 |
| 12 | `CXS_MAX_CREDIT` is at most 15 before Issue C, 63 from Issue C (skipped with `CXS_CREDIT_BEYOND_SPEC`) |
| 13 | Source/target ID widths are 0..8, user width 0..128 |
| 14 | `CXSSRCID` / `CXSTGTID` / `CXSUSER` need Issue D |

## 1.4 Streams and Interleaving

A CXS stream is the flits that share `CXSPRCLTYPE`, `CXSSRCID` and `CXSTGTID` (IHI 0079D 2.5.2). Flits of different streams may interleave (with `CXSCONTINUOUSDATA`, only where `CXSLAST` allows), so

packet framing is legal or not per stream: a flit may carry a START for stream B while stream A still has a packet open.

`cxs_cntl_track` keeps one packet-open bit per stream, selected by a key that `cxs_stream_key` builds from the low `CXS_STREAM_*_BITS` of the three fields. Tracking every possible stream would cost 2^19 bits, so the key bounds the state at 2^(P+S+T) bits. Streams that differ only above the key share one bit, so the key must cover every stream that can have a packet open at once. RAPIDS keys on the channel bits of `CXSSRCID`. The defaults (0/0/0) track the whole link as one stream, which is exactly the behaviour before 2026-10-16. Both ends of a link, and any monitor or snoop on it, must use the same key. With the key off, interleaved streams set `protocol_error`; `val/amba/test_cxs_stream_interleave.py` asserts both halves.

The framework's `CXSDepacketizer` and `CXSMonitor` track open packets per stream on the full spec key, so they need no parameter.

## 1.5 Credits Beyond the Spec

`CXS_MAX_CREDIT` is capped at 63 (15 before Issue C). `CXS_CREDIT_BEYOND_SPEC = 1` lifts the cap and nothing else: rule 12 is skipped, the credit counters widen to hold the pool, and every other rule still applies. It exists for a receiver whose buffer is a whole SRAM (the RAPIDS CXS tops default the pool to one channel's SRAM). Both ends of the link must be built with it; the framework's `CXSConfig(credit_beyond_spec=True)` mirrors it.

## 1.6 CXSCNTL Layout

`CXSCNTL` marks where packets start and end inside a flit. With N = `CXSMAXPKTPERFLIT`, the fields from bit 0 upward are:

```
START[N-1:0] | START0PTR .. START(N-1)PTR | END[N-1:0] | ENDERROR[N-1:0] | END0PTR .. END(N-1)PTR
```

- A start pointer is log2(W/128) bits with 16-byte alignment, or log2(W/32) with 4-byte alignment. The first byte of a packet is `STARTnPTR << 4` (or `<< 2`).
- An end pointer is always log2(W/32) bits and points at the packet's last aligned 4 bytes, so its last byte is `(ENDnPTR << 2) + 3`.
- `START` and `END` are thermometer codes, and `ENDERROR[n]` requires `END[n]`.

So `CXSCNTLWIDTH = N * (3 + start_ptr_w + end_ptr_w)`, and it is 0 when N = 1. `cxs_pkg::cntl_width()` computes it. The framework unit test `tests/unit/test_cxs_config.py` checks the formula against all 22 rows of IHI 0079D Tables 4-5 and 4-7, including each END field's bit position.

## 1.7 Link Control (CXS_LINK_CTRL = 1)

Two signals, CXSACTIVEREQ (driven by the transmitter) and CXSACTIVEACK (driven by the receiver), form a four-phase handshake. The link state is simply the pair of signal values; there is no separate state machine.

| {REQ, ACK} | State | Flits | Credits |
|---|---|---|---|
| 00 | STOP | none | all held by the receiver |
| 10 | ACTIVATE | none | the transmitter must accept early grants |
| 11 | RUN | yes | granted, spent, optionally returned |
| 01 | DEACTIVATE | in-flight only | the transmitter returns every credit; ACK falls only when all are home |

CXSDEACTHINT lets the receiver ask the transmitter to take the link down. Only the transmitter starts a state change. CXSACTIVEREQ may be captured asynchronously, so cxs_slave synchronizes it (ACTIVEREQ_SYNC_STAGES).

## 1.8 Implemented Signal Set

Every signal of IHI 0079D Table 3-1 and every check signal of Table 3-2 is implemented. An absent optional signal keeps a one-bit port, tied low on outputs and ignored on inputs, the same convention the AXIS endpoints use.

| Signal | Present when |
|---|---|
| CXSVALID, CXSDATA, CXSCRDGNT | always |
| CXSCNTL | CXS_MAX_PKT_PER_FLIT > 1 |
| CXSLAST / CXSPRCLTYPE | CXS_HAS_LAST / CXS_HAS_PRCLTYPE |
| CXSSRCID / CXSTGTID / CXSUSER | width > 0 |
| CXSCRDRTN, CXSACTIVEREQ, CXSACTIVEACK, CXSDEACTHINT | CXS_LINK_CTRL = 1 |
| *CHK (one per signal above, except | CXS_CHECK = 1 and the signal is present |

| Signal | Present when |
|---|---|
| CXSDEACTHINT) | |

## 1.9 Verification

The BFMs are the `cxs` family of the RDS-DV framework (`CocoTBFramework.components.cxs`): `CXSTransmitter`, `CXSReceiver`, `CXSMonitor`, `CXSConfig`, `CXSPacketizer` and `CXSDepacketizer`. The FUB side of each endpoint is driven by the framework GAXI BFMs. TB classes live in `bin/TBClasses/cxs/`.

| Test | DUT | Grid |
|---|---|---|
| `val/amba/test_cxs_master.py` | `cxs_master` | issues A-D, CXS-Lite, with and without link control and parity |
| `val/amba/test_cxs_slave.py` | `cxs_slave` | the same grid, each config also run with a 3-deep receive FIFO |
| `val/amba/test_cxs_master_cg.py` | `cxs_master_cg` | the same grid; must gate in STOP, never without link control |
| `val/amba/test_cxs_slave_cg.py` | `cxs_slave_cg` | the same grid; ACTIVEREQ must wake it |
| `val/amba/monitor-lite/test_cxs_{master,slave}_monlite.py` | the `_monlite` wrappers | the same grid; every Ocelus packet class checked |

```
source env_python
cd val/amba
make clean-all && make 'run-cxs_*-func-parallel'
```

Each traffic phase makes one claim: streaming, credit starvation (the DUT must stall, and never send without a credit), a full idle credit pool (exactly the bound, never more), idle and hint-driven deactivation with every credit returned, no deactivation inside a packet, and parity-error injection. The suite kills six deliberate RTL mutations: sending without a credit, never returning credits, granting one past the pool, dropping ACK with credits outstanding, ignoring `CXSDATACHK`, and deactivating mid-packet.

## 1.10 References


- Arm. *AMBA CXS Protocol Specification*, ARM IHI 0079, Issue A (Mar 2018), B (Dec 2020), C (Dec 2023), D (Sep 2025).

developer.arm.com/documentation/ihi0079. Chapters 2-7 throughout: 2.1.2 credit rules, 2.2 properties (Table 2-2) and compatibility (Table 2-3), 3.2 check signals (Table 3-2), 4.1 packet placement, 4.3 control fields (Tables 4-4, 4-5, 4-7), chapter 5 link activation, 6.1 continuous delivery, chapter 7 CXS-Lite. Citation only; the PDFs are not redistributed here.

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## 1.11 Navigation


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# 2 CXS Master

**Module:** `cxs_master.sv` **Location:** `rtl/amba/cxs/` **Status:** New (2026-10-15)

## 2.1 Overview

`cxs_master` is a CXS transmitter endpoint, the CXS counterpart of `axis4_master`. Flits arrive on a valid/ready FUB port, wait in a skid buffer, and leave on the credited CXS link one per cycle while the transmitter holds a credit and the link is in RUN. One parameter set covers every issue of IHI 0079 (see CXS Modules).

### 2.1.1 Key Features

- Credit counter for up to `CXS_MAX_CREDIT` credits; a grant is usable the cycle after it arrives (2.1.2)
- Link activation and deactivation (`Explicit_Credit_Return`): raises `CXSACTIVEREQ` on demand, returns every credit on `CXSCRDRTN` before STOP, never deactivates mid-packet
- Deactivation on idleness (`cfg_idle_cycles`), on `CXSDEACTHINT`, or on `cfg_link_enable` low
- `Odd_Byte_Parity` check-signal generation, with receiver check signals verified every cycle
- All optional signals: `CXSCNTL`, `CXSLAST`, `CXSPRCLTYPE`, `CXSSRCID`, `CXSTGTID`, `CXSUSER`
- No output depends combinationally on an input
- Sticky `protocol_error` and `parity_error` status

## 2.2 Parameters

The `CXS_*` property parameters are described in CXS Modules: Properties and Parameters. An illegal combination for `CXS_ISSUE` stops elaboration with the rule number.

| Parameter | Type | Default | Description |
|---|---|---|---|
| CXS_ISSUE | int | 4 (D) | Spec issue the property set is checked against |
| CXS_DATA_WIDT | int | 256 | `CXSDATAFLITWIDTH` |

| Parameter | Type | Default | Description |
|---|---|---|---|
| H | | | |
| CXS_MAX_PKT_PER_FLIT | int | 2 | `CXSMAXPKTPERFLIT`; 1 = CXS-Lite (no `CXSCNTL`) |
| CXS_START_ALIGN | int | 16 | `CXS_START_ALIGNMENT` in bytes |
| CXS_HAS_LAST / CXS_HAS_PRCLTYPE | bit | 0 | `CXS_LAST` / `CXS_PROTOCOL_TYPE` |
| CXS_SRCID_WIDTH / CXS_TGTID_WIDTH / CXS_USER_WIDTH | int | 0 | Issue D sideband widths, 0 = absent |
| CXS_MAX_CREDIT | int | 15 | Credits this transmitter can track |
| CXS_CREDIT_BEYOND_SPEC | bit | 0 | Lift the 63-credit cap (rule 12) for a whole-SRAM pool; both ends must agree |
| CXS_STREAM_PRCL_BITS / CXS_STREAM_SRCID_BITS / CXS_STREAM_TGTID_BITS | int | 0 | Per-stream packet framing key (see Streams and Interleaving); 0/0/0 = one stream |
| CXS_LINK_CTRL | bit | 1 | 1 = `Explicit_Credit_Return` |
| CXS_CHECK | bit | 0 | 1 = `Odd_Byte_Parity` |
| SKID_DEPTH | int | 4 | FUB skid buffer depth, 2..8 |

### 2.2.1 Derived Values (do not override)

| Name | Derivation | Meaning |
|---|---|---|
| CNTL_W | `cxs_pkg::cntl_width(N, DW, CXS_START_ALIGN)` | `CXSCNTLWIDTH`, 0 for CXS-Lite |
| CNTL_PW / SRCID_PW / TGTID_PW / USER_PW | max(width, 1) | Port widths of optional signals |
| DATACHK_W | DW/8 | `CXSDATACHK` width |
| CNTLCHK_W / USERCHK_W | max(ceil(width/8), 1) | `CXSCNTLCHK` / `CXSUSERCHK` widths |
| CRW | clog2(CXS_MAX_CREDIT + 1) | Credit counter width |

## 2.3 Ports

### 2.3.1 Clock, Reset and Configuration

| Port | Width | Direction | Description |
|---|---|---|---|
| aclk | 1 | Input | Clock |
| aresetn | 1 | Input | Active-low asynchronous reset |
| cfg_link_enable | 1 | Input | 0 = take the link down when idle and never raise it |
| cfg_link_hold | 1 | Input | 1 = keep the link in RUN with nothing to send |
| cfg_idle_cycles | 8 | Input | Idle RUN cycles before deactivating; 0 = never on idleness alone |

### 2.3.2 FUB Interface (valid/ready, input side)

| Port | Width | Direction | Description |
|---|---|---|---|
| fub_cxs_valid / fub_cxs_ready | 1 | In / Out | Flit handshake (ready = skid not full) |
| fub_cxs_data | DW | Input | Flit data; byte n is `data[8n+7:8n]` |
| fub_cxs_cntl | CNTL_PW | Input | `CXSCNTL`, passed through unchanged |
| fub_cxs_last | 1 | Input | `CXSLAST` |
| fub_cxs_prcltype | 3 | Input | `CXSPRCLTYPE` |
| fub_cxs_srcid / fub_cxs_tgtid / fub_cxs_user | *_PW | Input | Issue D sidebands |

### 2.3.3 CXS Transmitter Interface

| Port | Width | Direction | Description |
|---|---|---|---|
| m_cxs_valid | 1 | Output | `CXSVALID`: a flit this cycle (spends a credit) |
| m_cxs_data, m_cxs_cntl, m_cxs_last, m_cxs_prcltype, m_cxs_srcid, m_cxs_tgtid, m_cxs_user | as FUB | Output | Flit payload; zero while `CXSVALID` is low |
| m_cxs_crdgnt | 1 | Input | `CXSCRDGNT`: one credit per cycle |
| m_cxs_crdrtn | 1 | Output | `CXSCRDRTN`: a credit returned (only while `CXSACTIVEREQ` is low) |

| Port | Width | Direction | Description |
|---|---|---|---|
| m_cxs_activereq | 1 | Output | CXSACTIVEREQ |
| m_cxs_activeack | 1 | Input | CXSACTIVEACK, sampled into a flop |
| m_cxs_deacthint | 1 | Input | CXSDEACTHINT, sampled into a flop |

### 2.3.4 Check Signals (CXS_CHECK = 1)

| Port | Width | Direction | Description |
|---|---|---|---|
| m_cxs_validchk | 1 | Output | ~CXSVALID |
| m_cxs_datachk | DATACHK_W | Output | Odd parity, bit n covers data[8n+7:8n] |
| m_cxs_cntlchk | CNTLCHK_W | Output | Odd parity over CXSCNTL, short top byte |
| m_cxs_lastchk / m_cxs_prcltypechk / m_cxs_srcidchk / m_cxs_tgtidchk | 1 | Output | One bit each |
| m_cxs_userchk | USERCHK_W | Output | Odd parity over CXSUSER |
| m_cxs_crdrtnchk / m_cxs_activereqchk | 1 | Output | Inversions |
| m_cxs_crdgntchk / m_cxs_activeackchk | 1 | Input | Checked every cycle out of reset |

### 2.3.5 Status

| Port | Width | Direction | Description |
|---|---|---|---|
| busy | 1 | Output | Any FUB flit, skid entry, credit held, or link state other |

| Port | Width | Direction | Description |
|---|---|---|---|
| | | | than STOP |
| credits | CRW | Output | Credits held |
| link_state | 2 | Output | {ACTIVEREQ, sampled ACTIVEACK}: 00 STOP, 10 ACTIVATE, 11 RUN, 01 DEACTIVATE |
| in_packet | 1 | Output | A packet started on the link and not yet ended |
| protocol_error | 1 | Output | Sticky: a grant past `CXS_MAX_CREDIT`, ACK without REQ, or malformed `CXSCNTL` from the FUB |
| parity_error | 1 | Output | Sticky: `CXSCRDGNTCHK` or `CXSACTIVEACKCHK` mismatch |

## 2.4 Functional Description

### 2.4.1 Credits

`r_credits` updates every cycle by `+CXSCRDGNT - CXSVALID - CXSCRDRTN`. A gain and a spend in the same cycle cancel. A flit goes out when the skid holds one, the link is in RUN, and `r_credits` is nonzero. `r_credits` is a register, so a grant cannot be spent in the cycle it arrives. A grant that would push the count past `CXS_MAX_CREDIT` is dropped and flagged in `protocol_error`.

### 2.4.2 Link control

| From | To | When |
|---|---|---|
| STOP | ACTIVATE | `cfg_link_enable` and (a FUB flit, a skid entry, or `cfg_link_hold`) |
| ACTIVATE | RUN | the receiver raises `CXSACTIVEACK` |
| RUN | DEACTIVATE | skid empty, no packet open, and one of: `cfg_link_enable` low, `CXSDEACTHINT` with nothing to send, `cfg_idle_cycles` idle cycles |
| DEACTIVATE | STOP | the receiver drops `CXSACTIVEACK` after every credit is home |

While `CXSACTIVEREQ` is low, `CXSCRDRTN` returns one held credit per cycle, including grants that arrive late. `CXSVALID` needs `CXSACTIVEREQ`, so the two are never high together.

### 2.4.3 Packet tracking

`cxs_cntl_track` counts the packets each sent flit starts and ends. That gives `in_packet`, which blocks deactivation, and format checks: thermometer `START`/`END`, `ENDERROR` only with `END`, and no packet ending before it started.

---

## 2.5 Timing Characteristics

- FUB to link: one skid buffer, so a flit accepted at cycle t can be sent at t+1 at the earliest.
- `CXSCRDGNT` at cycle t: usable at t+1.
- `CXSACTIVEACK` rising at cycle t: the first flit goes at t+1 at the earliest (registered sample).
- Back-to-back flits at one per cycle need `CXS_MAX_CREDIT` at least as large as the credit round trip (2.1.2).

---

## 2.6 Usage Example

```
cxs_master #(
    .CXS_ISSUE            (cxs_pkg::CXS_ISSUE_B),
    .CXS_DATA_WIDTH       (512),
    .CXS_MAX_PKT_PER_FLIT (2),
    .CXS_HAS_LAST         (1'b1),
    .CXS_MAX_CREDIT       (15),
    .CXS_LINK_CTRL        (1'b1),
    .CXS_CHECK            (1'b0)
) u_cxs_tx (
    .aclk            (aclk),
    .aresetn         (aresetn),
    .cfg_link_enable (1'b1),
    .cfg_link_hold   (1'b0),
    .cfg_idle_cycles (8'd32),
    .fub_cxs_valid   (flit_valid),   .fub_cxs_ready   (flit_ready),
    .fub_cxs_data    (flit_data),    .fub_cxs_cntl    (flit_cntl),
    .fub_cxs_last    (flit_last),    .fub_cxs_prcltype(3'b000),
    .fub_cxs_srcid   ('0), .fub_cxs_tgtid ('0), .fub_cxs_user ('0),
    .m_cxs_valid     (CXSTXVALID),   .m_cxs_data      (CXSTXDATA),
    .m_cxs_cntl      (CXSTXCNTL),    .m_cxs_last      (CXSTXLAST),
    .m_cxs_crdgnt    (CXSTXCRDGNT),  .m_cxs_crdrtn    (CXSTXCRDRTN),
    .m_cxs_activereq (CXSTXACTIVEREQ), .m_cxs_activeack
(CXSTXACTIVEACK),
    .m_cxs_deacthint (CXSTXDEACTHINT),
    .m_cxs_crdgntchk ('0), .m_cxs_activeackchk ('0),
    // remaining outputs left unconnected or monitored
    .busy (), .credits (), .link_state (), .in_packet (),
    .protocol_error (tx_proto_err), .parity_error ()
);
```

## 2.7 Design Notes

- **The FUB supplies `CXSCNTL`.** This block carries flits; it does not pack packets into them. The framework `CXSPacketizer` shows the placement rules.
- **CXS-Lite** (`CXS_MAX_PKT_PER_FLIT = 1`, Issue C/D): there is no `CXSCNTL`, every flit is one packet, and `in_packet` is always 0.
- **Check signals are generated, not carried.** Protection starts at the interface, so the payload `*CHK` bits are computed from the outgoing flit. That differs from `axis5_master`, which carries the FUB's check bits end to end.

## 2.8 Related Modules

- **CXS Slave**: the receiver
- **CXS Master CG**: clock-gated variant
- **AXIS4 Master**: the valid/ready endpoint this one mirrors
- `gaxi_skid_buffer`, `cxs_cntl_track`, `cxs_odd_byte_parity`, `cxs_pkg`

---

## 2.9 Testing

`val/amba/test_cxs_master.py` uses a GAXI master on `fub_cxs_*`, the framework `CXSReceiver` on `m_cxs_*`, and an independent `CXSMonitor`. The phases are fast, starve, gappy, back-to-back, idle pool fill, idle deactivation, reactivation, hint deactivation, mid-packet idle and parity injection.

```
source env_python
cd val/amba && make clean-all && make run-cxs_master-func-parallel
```

## 2.10 References

- Arm. *AMBA CXS Protocol Specification*, ARM IHI 0079D, Sep. 2025 (and Issues A-C). developer.arm.com/documentation/ihi0079/d: 2.1.1 clock and reset, 2.1.2 credit exchange, 3.2 check signals, 5.2-5.6 link activation, 6.1 continuous delivery. Citation only.

---

## 2.11 Navigation

- **Back to CXS Index**
- **Back to rtl-amba Index**
- **Back to Main Documentation Index**

# 3 CXS Master with Clock Gating

**Module:** `cxs_master_cg.sv` **Location:** `rtl/amba/cxs/` **Status:** New (2026-10-15)

## 3.1 Overview

`cxs_master_cg` puts cxs_master behind one `amba_clock_gate_ctrl`. It has the same parameters and ports as the core, minus `busy`, plus the clock-gating set: `CG_IDLE_COUNT_WIDTH`, `cfg_cg_enable`, `cfg_cg_idle_count`, `cg_gating` and `cg_idle`.

## 3.2 When It Gates

A credited link changes state with no handshake the gated side could stall. A `CXSCRDGNT` that arrives while the clock is stopped is a credit lost for good. The core's `busy` therefore covers every state in which the receiver may still grant: `CXSACTIVEREQ` or the sampled `CXSACTIVEACK` high, or any credit held. As a result:

| CXS_LINK_CTRL | Gates? | Why |
|---|---|---|
| 1 | Only in STOP with zero credits | The receiver "must not send credit" in STOP (IHI 0079D 5.2). A FUB flit wakes the block, which then raises `CXSACTIVEREQ` |
| 0 | Never | The receiver grants whenever it has room, so the core holds credits indefinitely and `busy` never falls |

The second row is a property of the protocol, not a shortcoming of this wrapper. To save clock power, use link control.

## 3.3 Activity Terms

The terms follow `vault/handbook/design/clock-gating-activity-terms.md`:

| Term | Expression | Reason |
|---|---|---|
| user_valid | `fub_cxs_valid \| busy` | FUB VALID plus all pending work |

| Term | Expression | Reason |
|---|---|---|
| axi_valid | `m_cxs_crdgnt \| m_cxs_activeack` | The peer's VALID-like inputs, synchronous to `aclk` (5.5) |

No READY appears in either term; CXS has none. The event outputs `CXSVALID` and `CXSCRDRTN` are masked with `!cg_gating` for the wake-latency overlap, and their check bits follow the masked value (a masked 0 carries check 1). `fub_cxs_ready` is forced low while gated.

## 3.4 Testing

`val/amba/test_cxs_master_cg.py` runs the master traffic with `cfg_cg_enable = 1` and checks every flit. It then makes two gating claims: with link control the block **must** gate during an idle window in STOP, and without link control it must **never** gate.

## 3.5 Navigation

- **Back to CXS Index**
- **CXS Master**

# 4 CXS Slave

**Module:** `cxs_slave.sv` **Location:** `rtl/amba/cxs/` **Status:** New (2026-10-15)

## 4.1 Overview

`cxs_slave` is a CXS receiver endpoint, the CXS counterpart of `axis4_slave`. Flits arrive on the credited link, land in a receive FIFO, and leave on a valid/ready FUB port. The receive FIFO **is** the credit pool. A credit is granted only when a FIFO slot is free to back it, so the receiver never refuses a flit. That is the guarantee 2.1.2 requires ("the Receiver must guarantee that it can receive one flit of data for each credit that it grants").

### 4.1.1 Key Features

- Credit grant bounded by FIFO room and `CXS_MAX_CREDIT`; a spent or returned credit is reissued the next cycle at the earliest
- `RX_FIFO_DEPTH` may be smaller than `CXS_MAX_CREDIT`, in which case the FIFO sets the pool size
- Link control: synchronizes the asynchronous `CXSACTIVEREQ`, raises `CXSACTIVEACK`, grants only in RUN, and drops ACK only once every credit is home
- `CXSDEACTHINT` driven from `cfg_deact_hint`
- `Odd_Byte_Parity` checking of every received signal, with its own check signals generated
- Sticky `protocol_error`: a flit or return with no credit, `CXSVALID` together with `CXSCRDRTN`, FIFO overflow, malformed `CXSCNTL`

## 4.2 Parameters

The `CXS_*` property parameters are the same as for cxs_master. Two more:

| Parameter | Type | Default | Description |
|---|---|---|---|
| RX_FIFO_DEPTH | int | CXS_MAX_CREDIT | Receive FIFO depth, i.e. the credit pool. Any depth >= 1. Below the credit round trip it throttles the |

| Parameter | Type | Default | Description |
|---|---|---|---|
| | | | link |
| CXS_CREDIT_BEYOND_SPEC | bit | 0 | Lift the 63-credit cap (rule 12) for a whole-SRAM pool; both ends must agree |
| CXS_STREAM_PRCL_BITS / CXS_STREAM_SRCID_BITS / CXS_STREAM_TGTID_BITS | int | 0 | Per-stream packet framing key (see Streams and Interleaving); 0/0/0 = one stream |
| ACTIVEREQ_SYNC_STAGES | int | 2 | Synchronizer flops on `CXSACTIVEREQ` (and its check bit); 0 = known-synchronous system |

Derived values (`CNTL_W`, the `*_PW` port widths, the check widths and `CRW`) are as for `cxs_master`.

---

## 4.3 Ports

### 4.3.1 Clock, Reset and Configuration

| Port | Width | Direction | Description |
|---|---|---|---|
| aclk / aresetn | 1 | Input | Clock, active-low asynchronous reset |
| cfg_rx_enable | 1 | Input | 0 = never activate, never grant |
| cfg_deact_hint | 1 | Input | Drives |

| Port | Width | Direction | Description |
|---|---|---|---|
| | | | CXSDEACTHINT (registered) |

### 4.3.2 CXS Receiver Interface

| Port | Width | Direction | Description |
|---|---|---|---|
| s_cxs_valid | 1 | Input | CXSVALID |
| s_cxs_data, s_cxs_cntl, s_cxs_last, s_cxs_prcltype, s_cxs_srcid, s_cxs_tgtid, s_cxs_user | see cxs_master | Input | Flit payload; absent signals ignored |
| s_cxs_crdgnt | 1 | Output | CXSCRDGNT, registered |
| s_cxs_crdrtn | 1 | Input | CXSCRDRTN |
| s_cxs_activereq | 1 | Input | CXSACTIVEREQ, asynchronous, synchronized here |
| s_cxs_activeack | 1 | Output | CXSACTIVEACK |
| s_cxs_deacthint | 1 | Output | CXSDEACTHINT |

### 4.3.3 Check Signals (CXS_CHECK = 1)

The s_cxs_*chk inputs mirror the m_cxs_*chk outputs of cxs_master. s_cxs_crdgntchk and s_cxs_activeackchk are generated outputs.

### 4.3.4 FUB Interface (valid/ready, output side)

| Port | Width | Direction | Description |
|---|---|---|---|
| fub_cxs_valid / fub_cxs_ready | 1 | Out / In | FIFO not empty / consumer ready |
| fub_cxs_data … fub_cxs_user | as received | Output | The flit, absent fields zero |

### 4.3.5 Status

| Port | Width | Direction | Description |
|---|---|---|---|
| busy | 1 | Output | A flit arriving or queued, credits out, or link state other than STOP |
| credits_outstanding | CRW | Output | Granted and not yet spent or returned |
| link_state | 2 | Output | {synchronized ACTIVEREQ, ACTIVEACK} |
| in_packet | 1 | Output | A packet started and not yet ended |
| protocol_error / parity_error | 1 | Output | Sticky error flags |

## 4.4 Functional Description

### 4.4.1 Credit pool

`r_out` counts credits outstanding, including the grant on the wire this cycle. Each cycle the block works out the state after the clock edge, before any new grant:

```
out_next = r_out + CXSCRDGNT(now) - CXSVALID - CXSCRDRTN
occ_next = fifo_count + CXSVALID - fub_pop
grant    = RUN && out_next < CXS_MAX_CREDIT && occ_next + out_next < RX_FIFO_DEPTH
```

`r_crdgnt <= grant`, so `CXSCRDGNT` is a flop fed from flops and this cycle's handshake. A credit consumed at cycle t is reissued at t+1 at the earliest.

`gaxi_fifo_sync` cannot be built 1 deep, but one credit with a one-flit buffer is a legal receiver. The FIFO is therefore built at least 2 deep while the grant still uses `RX_FIFO_DEPTH`, so the spare slot is never granted against.

### 4.4.2 Link control

`CXSACTIVEREQ` and `CXSACTIVEREQCHK` cross through one `glitch_free_n_dff_arn`. The check bit crosses **inverted**: the synchronizer flops reset to 0, and the correct check for a reset-low

ACTIVEREQ is 1. Carried uninverted, the pair would read `{0,0}`, a parity error, for every cycle the synchronizer is flushing after reset. The test suite caught exactly that.

- ACTIVATE to RUN: `CXSACTIVEACK` rises when the synchronized ACTIVEREQ is high and `cfg_rx_enable` is set.
- Credits are granted only while both are high.
- DEACTIVATE: flits still in flight are accepted, and ACK falls only when `out_next` is 0.

---

## 4.5 Timing Characteristics

- Link to FUB: the FIFO is in mux-read mode, so a flit received at cycle t is offered at t+1.
- Credit reuse: a slot freed at cycle t (FUB pop) supports a grant driven at t+1.
- ACTIVEREQ to ACTIVEACK: `ACTIVEREQ_SYNC_STAGES` + 1 cycles; the first grant follows one cycle after ACK.

---

## 4.6 Usage Example

```
cxs_slave #(
    .CXS_ISSUE            (cxs_pkg::CXS_ISSUE_D),
    .CXS_DATA_WIDTH       (256),
    .CXS_MAX_PKT_PER_FLIT (3),
    .CXS_START_ALIGN      (4),
    .CXS_SRCID_WIDTH      (4),
    .CXS_TGTID_WIDTH      (2),
    .CXS_MAX_CREDIT       (15),
    .RX_FIFO_DEPTH        (8)
) u_cxs_rx (
    .aclk (aclk), .aresetn (aresetn),
    .cfg_rx_enable (1'b1), .cfg_deact_hint (want_idle),
    .s_cxs_valid (CXSRXVALID), .s_cxs_data (CXSRXDATA), .s_cxs_cntl
(CXSRXCNTL),
    .s_cxs_last ('0), .s_cxs_prcltype ('0),
    .s_cxs_srcid (CXSRXSRCID), .s_cxs_tgtid (CXSRXTGTID), .s_cxs_user
('0),
    .s_cxs_crdgnt (CXSRXCRDGNT), .s_cxs_crdrtn (CXSRXCRDRTN),
    .s_cxs_activereq (CXSRXACTIVEREQ), .s_cxs_activeack
(CXSRXACTIVEACK),
    .s_cxs_deacthint (CXSRXDEACTHINT),
    // check inputs tied when CXS_CHECK = 0
    .s_cxs_validchk ('0), .s_cxs_datachk ('0), .s_cxs_cntlchk
```

```
('0), .s_cxs_lastchk ('0),
    .s_cxs_prcltypechk ('0), .s_cxs_srcidchk ('0), .s_cxs_tgtidchk
('0), .s_cxs_userchk ('0),
    .s_cxs_crdrtnchk ('0), .s_cxs_activereqchk ('0),
    .s_cxs_crdgntchk (), .s_cxs_activeackchk (),
    .fub_cxs_valid (rx_valid), .fub_cxs_ready
(rx_ready), .fub_cxs_data (rx_data),
    .fub_cxs_cntl (rx_cntl), .fub_cxs_last (), .fub_cxs_prcltype (),
    .fub_cxs_srcid (rx_src), .fub_cxs_tgtid (rx_tgt), .fub_cxs_user
(),
    .busy (), .credits_outstanding (), .link_state (), .in_packet (),
    .protocol_error (rx_proto_err), .parity_error ()
);
```

## 4.7 Related Modules

- **CXS Master**: the transmitter
- **CXS Slave CG**: clock-gated variant
- **AXIS4 Slave**: the valid/ready endpoint this one mirrors
- `gaxi_fifo_sync`, `glitch_free_n_dff_arn`, `cxs_cntl_track`, `cxs_odd_byte_parity`

## 4.8 Testing

`val/amba/test_cxs_slave.py` uses the framework `CXSTransmitter` on `s_cxs_*`, a GAXI slave on `fub_cxs_*`, and a `CXSMonitor`. Every config runs twice, once with `RX_FIFO_DEPTH = CXS_MAX_CREDIT` and once with a 3-deep FIFO. The idle pool-fill phase requires exactly `min(RX_FIFO_DEPTH, CXS_MAX_CREDIT)` credits at the transmitter.

```
source env_python
cd val/amba && make clean-all && make run-cxs_slave-func-parallel
```

## 4.9 References


- Arm. *AMBA CXS Protocol Specification*, ARM IHI 0079D, Sep. 2025 (and Issues A-C). developer.arm.com/documentation/ihi0079/d: 2.1.2 credit rules, 5.2 receiver obligations in each state, 5.4 race conditions, 5.5 asynchronous `CXSACTIVEREQ`. Citation only.

## 4.10 Navigation

- **Back to CXS Index**
- **Back to rtl-amba Index**
- **Back to Main Documentation Index**

# 5 CXS Slave with Clock Gating

**Module:** `cxs_slave_cg.sv` **Location:** `rtl/amba/cxs/` **Status:** New (2026-10-15)

## 5.1 Overview

`cxs_slave_cg` puts cxs_slave behind one `amba_clock_gate_ctrl`. It has the same parameters and ports as the core, minus `busy`, plus `CG_IDLE_COUNT_WIDTH`, `cfg_cg_enable`, `cfg_cg_idle_count`, `cg_gating` and `cg_idle`.

IHI 0079D 5.5 anticipates this block: “It is common for the clock in the Receiver to stop during the STOP state … The assertion of the CXSACTIVEREQ signal might be used to restart that clock.”

## 5.2 When It Gates

The core’s `busy` covers every state in which a flit or credit return can still arrive: ACK or the synchronized ACTIVEREQ high, any credit outstanding, or a flit waiting on the FUB side.

| CXS_LINK_CTRL | Gates? | Why |
|---|---|---|
| 1 | Only in STOP, every credit home, FIFO empty | The transmitter “must not send flits or credit returns” in STOP (5.2); a rising `CXSACTIVEREQ` wakes the block |
| 0 | Never | Credits stay outstanding indefinitely, so a flit can arrive on any cycle |

## 5.3 Activity Terms

| Term | Expression | Reason |
|---|---|---|
| user_valid | `busy` | Includes `fub_cxs_valid`, the pending output |
| axi_valid | `s_cxs_valid \| s_cxs_crdrtn \| ACTIVEREQ (synchronized)` | Peer VALIDs and the wake request |

`CXSACTIVEREQ` is asynchronous, so the wake term passes through its own `glitch_free_n_dff_arn` on the **free-running** `aclk`. A wake term must be in the gate’s own

clock domain, and the core's synchronizer sits on the gated clock and is frozen in STOP. `CXSCRDGNT` (and its check bit) and `fub_cxs_valid` are masked with `!cg_gating`.

## 5.4 Testing

`val/amba/test_cxs_slave_cg.py` streams flits, takes the link to STOP, idles, and requires the block to have gated during that window. It then sends a wake phase that must be delivered intact, which proves `CXSACTIVEREQ` restarted the clock. Without link control it must never gate.

## 5.5 Navigation

- **Back to CXS Index**
- **CXS Slave**